

HOUJIANG LIU

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EDUCATION

University of Texas at Austin, School of Information, 2021– Expected May 2027

Ph.D. Candidate of Information Science

From Automation to Agency: Evaluating Computational Support for Human Research Ideation

Committee: Matthew Lease (advisor), Jacek Gwizdka, Ahmer Arif (Barnard College, Columbia University), and Aniket Kittur (Carnegie Mellon University)

Northeastern University, College of Arts, Media and Design, June 2020

M.F.A. in Experience Design

Enriching Gifts, Experience Design to Understand Human Relationships

Committee: Miso Kim (advisor), Kristian Kloeckl, & Meryl Alper

Jilin University, College of Physics, July 2015

B.S. in Applied Physics

RESEARCH PROFILE

My research lies at the intersection of **Human-Computer Interaction**, **Design**, and **Responsible AI**, with a focus on how AI should be designed to support rather than erode human expertise. I study how increasingly capable AI systems reshape the intellectual work of experts, particularly scientists, journalists, and other knowledge workers, examining how automation changes human effort, creativity, agency, and professional judgment. Through empirical studies and interactive system design, my research develops human-centered principles, methods, and systems for responsible human–AI collaboration that preserve meaningful human involvement while enabling AI to extend expert capabilities.

HONORS, AWARDS & FUNDING

Research Funding & Fellowships

- **Research Grant Award**, Cosmos Institute, 2026 — \$5,000, AI and Human Autonomy
- **E. Glynn Harmon Excellence Fund**, University of Texas at Austin, 2026 — \$1,000
- **Outstanding Research Fellowship**, University of Texas at Austin, 2025-26 — \$38,000
- **Summer Research Fellowship**, University of Texas at Austin, 2023 — \$10,832
- **Summer Research Fellowship**, Northeastern University, 2019 — \$5,000
- **Graduate Scholarship**, Northeastern University, 2018–20 — 50% tuition remission
- **Undergraduate Merit Scholarship**, Jilin University, 2012–13, 2015 — with direct admission to graduate program

Honors & Awards

- **Best Paper Honorable Mention**, ACM CSCW, 2024 — top 4% of submitted papers
- **Professional Development Award**, University of Texas at Austin, 2023, 2024, 2025

- **Information is Beautiful Awards** — Longlist, Data Visualization Society, 2022
- **Dean's Project Award for Research, Scholarship, and Creative Activity**, Northeastern University, 2020 — \$2,000
- **First Design Award**, Health x Design Challenge, Mad*Pow, 2019 — \$5,000
- **Innovative Design Award**, Reinventing Tencent Map, Tencent, 2014
- **Second Prize**, Math Modeling, Jilin University, 2012

PUBLICATIONS (Authors with equal contributions are marked with *)

Manuscripts Under Review [U]

- [U3] **Houjiang Liu**, Yiheng Su, and Matthew Lease.
 “Rhetorical XAI: A Design Menu for AI Explanations.” Under review, *ACM CHI 2027* [[arXiv](#)]
- [U2] **Houjiang Liu***, Yujin Choi*, Sanjana Gautam, Gabriel Jaffe, Soo Young Rieh, and Matthew Lease.
 “Who Owns Creativity and Who Does the Work? Trade-offs in LLM-Supported Research Ideation. Under review, *International Journal of Human-computer Interaction* [[arXiv](#)]
- [U1] **Houjiang Liu***, Li Shi*, Yian Wong, Utkarsh Mujumdar, Dan Zhang, Jacek Gwizdka, and Matthew Lease.
 “Reducing Confirmation Bias on Controversial Issues through LLM-Generated Multi-Persona Debates.” Minor revision resubmitted, *Information Processing & Management* [[arXiv](#)]

Peer-Reviewed Conference Proceedings and Journal [P]

- [P12] Sanjana Gautam, **Houjiang Liu**, Yujin Choi, and Matthew Lease.
 “How Researchers Navigate Accountability, Transparency, and Trust When Using AI Tools in Early-Stage Research: A Think-Aloud Study.” *Proceedings of the 9th ACM Conference on Fairness, Accountability, and Transparency*, pp. 1494–1512, 2026, Montréal, Canada, 2026. [[DOI](#), [arXiv](#)]
- [P11] **Houjiang Liu**, Jacek Gwizdka, and Matthew Lease.
 “Exploring Multidimensional Checkworthiness: Designing AI-assisted Claim Prioritization for Human Fact-checkers.” *Proceedings of ACM Human-Computer Interaction*, 9 (7): CSCW292, 49 pages, 2025. [[DOI](#), [arXiv](#), [Code](#), [Slides](#), [Video](#)]
- [P10] **Houjiang Liu***, Anubrata Das*, Alexander Boltz*, Didi Zhou, Daisy Pinaroc, Matthew Lease, and Min Kyung Lee.
 “Human-centered NLP Fact-checking: Co-Designing with Fact-checkers using Matchmaking for AI.” *Proceedings of ACM Human-Computer Interaction*. 8, CSCW2, Article 423, 44 pages, 2024. [[DOI](#), [arXiv](#), [Slides](#)]
Best Paper Honorable Mention (top 4%)
- [P9] Anubrata Das, **Houjiang Liu**, Venelin Kovatchev, and Matthew Lease.
 “The state of human-centered NLP technology for fact-checking.” *Information Processing & Management*, 60(2) :103219, 2023. [[DOI](#), [arXiv](#)]

- [P8] **Houjiang Liu***, Estefania Ciliotta Chehade*, William Albert, and Miso Kim.
 “An Empirical Study of How Service Designers Use Metrics.” *She Ji: The Journal of Design, Economics, and Innovation*, 9(3) :402–427, 2023. [[DOI](#), [Blog](#)]
- [P7] Mingzhu Li, Miso Kim, Liqing Huang, and **Houjiang Liu**.
 “Designing for people you can’t meet: Persuasive design to nurture seniors’ participation in community services during the COVID-19 pandemic.” *The Design Journal*, 26(6) :934–955, 2023. [[DOI](#)]
- [P6] Sara Colombo, Estefania Ciliotta Chehade, Lucia Marengo, **Houjiang Liu**, Piero Molino, and Paolo Ciuccarelli.
 “Design for Emergency: How Digital Technologies Enabled an Open Design Platform to Respond to COVID-19.” *Interacting with Computers*, 35(5) :650–665, 2023. [[DOI](#), [Website](#), [Blog](#), [Video](#)]
- [P5] **Houjiang Liu** and Miso Kim.
 “Is gift a token of gratitude or an imposition of identity? Facilitating positive consequences of gift-giving with receiver-centred design.” *Proceedings of the 10th International Association of Societies of Design Research (IASDR)*, Milan, Italy, 2023. [[DOI](#), [Slides](#)]
- [P4] Jiabao Li, **Houjiang Liu**, Jilie Zeng, Di Wu, Ying Ding, and Alec McGail.
 “Shifting Winds: Gendered Structures of Academic Mentorship.” *Proceedings of 2022 IEEE VIS Arts Program (VISAP)*, pp. 14–29, ieeexplore, Oklahoma City, USA, 2022. [[DOI](#), [Website](#), [Code](#)]
- [P3] **Houjiang Liu**, Ray Lc, Miso Kim, Concettina Cormio, and Meixin Yu.
 “Designing for Distance Nursing: Reconnecting Nursing Students with Senior Home Residents During COVID-19.” *Proceedings of the 9th International Association of Societies of Design Research (IASDR)*, Hong Kong, 2022. [[DOI](#), [Slides](#)]
- [P2] **Houjiang Liu**, Miso Kim, Mingzhu Li, and Shruthi Lakshmi Narayan.
 “Technology, Autonomy, and Participation: Designing Community Games and Services to Enhance Older Adults’ Technology Literacy.” *Proceedings of the 9th International Association of Societies of Design Research (IASDR)*, Hong Kong, 2022. [[DOI](#), [Slides](#)]
- [P1] **Houjiang Liu**, Miso Kim, Canqun He, and Tia Thomson.
 “Autonomy as a Design Principle: Service Design for the Technology Literacy of Older Adults.” *Proceedings of the Cumulus Conference on Design Culture(s)*, Rome, Italy, 2021. [[DOI](#)]

Lightly Reviewed Extended Abstracts, Short Papers, & Posters [E]

- [E4] Huimin Xu, **Houjiang Liu**, Yan Leng, and Ying Ding.
 “Adapting to LLMs: How Insiders and Outsiders Reshape Scientific Knowledge Production.” *International Conference on the Science of Science and Innovation*, Boulder, USA, 2026. [[arXiv](#)]
- [E3] **Houjiang Liu***, Yujin Choi*, Sanjana Gautam, Gabriel Jaffe, Soo Young Rieh, and Matthew Lease.
 “Effects of Varying Control vs. Automation in Human-in-the-loop Research

Ideation." *Amazon Research Day, Workshop on AI co-scientist: Accelerating Scientific Discovery through Intelligent AI Collaboration*, Palo Alto, USA, 2026. [[pdf](#)]

- [E2] Anubrata Das, **Houjiang Liu**, Venelin Kovatchev, and Matthew Lease. "The Need for Human-centered Design in Fact-checking Research." *Proceedings of the 1st Information Processing & Management (IP&M) Conference*, Xiamen, China, 2022. [[pdf](#)]
- [E1] **Houjiang Liu**. "Enriching Family Gifts: Reciprocity in Family Relationships." (Extended Abstract of C5). *NA PhD by Design Symposium*, New York University, USA, 2020. (Symposium canceled due to COVID19 shutdown)

Preprints [M]

- [M1] Jiaxin An, Siqu Yi, Yao Lyu, **Houjiang Liu**, and Yan Zhang. "Conversational Agents for Older Adults' Health: A Systematic Literature Review." 2025. [[arXiv](#)]

SELECTED TALKS, PRESENTATIONS & EXHIBITIONS

Invited Talks

- "Assessing Creativity vs. Efficiency Tradeoffs across Control Levels of LLM-Supported Research Ideation." *CosmicAI Biweekly Seminar*, February 2026.
- "Exploring Multidimensional Checkworthiness: Designing AI-assisted Claim Prioritization for Human Fact-checkers." *iSchools Global Doctoral Seminar Series*, August 2025.
- "Generating Multi-Persona Debates with LLMs to Explore Diverse Perspectives on Controversial Topics." *Kungfu.AI Guest Lecture*, May 2025.

Selected Presentations

- "Generating Multi-Persona Debates with LLMs to Explore Diverse Perspectives on Controversial Topics."
 - o *Good Systems Annual Symposium, Shaping the Future of Ethical AI*. University of Texas at Austin, Spring 2025.
 - o *iSchool AI Showcase: Community-Engaged Research on AI and Society*. University of Texas at Austin, Fall 2024.
- "An Exploration of Service Design Metrics." *Research, Innovation, Scholarship, Entrepreneurship (RISE), Virtual EXPO*. Northeastern University, 2021.
- "Participatory Design for Senior Healthcare: Service Design to Improve Seniors' Health Literacy and Autonomy." *Research, Innovation, Scholarship, Entrepreneurship (RISE)*. Northeastern University, 2020.

Selected Design Exhibitions

- Biophilia: Mapping the Emotional Pulse of Urban Nature. *Information+ Conference*, Boston, 11/14-16, 2025 [[Exhibition](#)]
- Shifting Winds: Gendered Structures of Academic Mentorship. *IEEE VIS Arts Program (VISAP)*, Oklahoma City, 10/16-21, 2022 [[Exhibition](#)]
- Data Sonification Archive. 2022 *Design Research Week*, Boston, 4/25-29 [[Website](#)]

- How Food is Served in Early Care Center. *Northeastern University Visualization Consortium (NUVis) Demofest*. Boston, 2019.

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant, AI & Human-Centered Computing Group, University of Texas at Austin. 2021–present

Conduct interdisciplinary research on human–AI interaction, with a focus on creativity support, expert work, and responsible AI. Lead research activities across funded projects, including empirical study design, interactive system prototyping, data analysis, manuscript development, and project coordination.

Selected funded projects: UT Good Systems; CosmicAI NSF/Simons-supported research institute.

Design Researcher, Center for Design, Northeastern University. 2020–21

Conducted design research in service design and information visualization, applying qualitative and participatory design methods to investigate technology use and needs among older adults.

Selected funded projects: Center for Design Bridges Fellowship

Lecturer, College of Arts, Media and Design, Northeastern University. 2020–21

Taught interaction design and user experience courses, with an emphasis on design research, prototyping, and human-centered design methods.

User Experience Designer, SeetaTech, Beijing. 2017–18

Designed user experiences for AI-powered products, translating user research and design insights into product strategies, interaction flows, and interface designs in collaboration with product and engineering teams.

Interaction Designer, JD.com, Beijing. 2015–2017

Designed user experiences for large-scale e-commerce products across web and mobile platforms, conducting user research and developing interaction frameworks in collaboration with product and engineering teams.

Interaction Design Intern (2015 Spring, JD.com; 2013 Summer, Lenovo; 2012 Summer, Midea Group)

TEACHING EXPERIENCE

Teaching Fellow, College of Arts, Media and Design, Northeastern University

ARTG2400 Interaction Design, Spring 2021 – Instructor of Record

ARTG2401 Interaction Design Tools, Fall 2019, Spring 2020 – Instructor of Record

Developed and taught coursework in interaction design, design research, prototyping, and human-centered design methods.

Teaching Assistant, College of Arts, Media and Design, Northeastern University

ARTG6330 Info Design Mapping Strategies, Fall 2020

ARTG6900 Design for Dignity, Fall 2019

SERVICE

Selected Reviewing

(5 x Recognized for Outstanding Reviews)

- ACM CHI, 2026 (1 outstanding review)
- ACM CSCW, 2022–27 (3 outstanding review)
- ACM UIST, 2024, 2026 (1 outstanding review)
- ACM Collective Intelligence (CI), 2025
- ACM DIS, 2025
- *International Journal of Human-Computer Interaction* 2026
- *International Journal of Intercultural Relations* 2026
- *Information Processing & Management* 2026
- Design Research Society Conference (DRS) 2024

Student Volunteer

- ACM CSCW, 2024-25
- ACM International World Wide Web Conference (WebConf), 2023
- ACM SIGIR Conference on Human Information Interaction Retrieval (CHIIR), 2022
- User Experience Professionals Association Conference (UXPA) International, 2015

Ph.D. Representative, School of Information, University of Texas at Austin

- Research Committee 2024–26
- Master Studies Committee 2021–22